

CONSULTING PROPOSAL

Diabeties Check Via Report Smart Ai MI Webapp

Industry:	Healthcare
Duration:	6 months
Users:	500+
Tech Stack:	python, nextjs, machinelearning
Date:	February 27, 2026

1. Project Overview

Parameter	Details
Project Title	Diabeties Check Via Report Smart Ai MI Webapp
Client	-
Industry	Healthcare
Duration	6 months
Expected Users	500
Tech Stack	python, nextjs, machinelearning

2. Executive Summary

The Diabetes Check via Report Smart AI/ML Webapp is a clinical decision support system that analyzes patient health reports to predict diabetes risk levels and provide actionable insights. It addresses the critical Healthcare industry challenge of early diabetes detection, enabling providers to intervene before complications develop and reducing hospital readmission rates. The system is projected to reduce diagnosis processing time by 40% and decrease unnecessary specialist referrals by 25%, directly improving patient outcomes while optimizing resource utilization. Primary beneficiaries include primary care clinics, endocrinologists, managed care organizations, and public health agencies implementing preventive care programs. The solution delivers strategic ROI through reduced long-term complication costs and ensures HIPAA/GDPR compliance with audit-ready architecture suitable for 2026 regulatory requirements.

3. Technical Approach

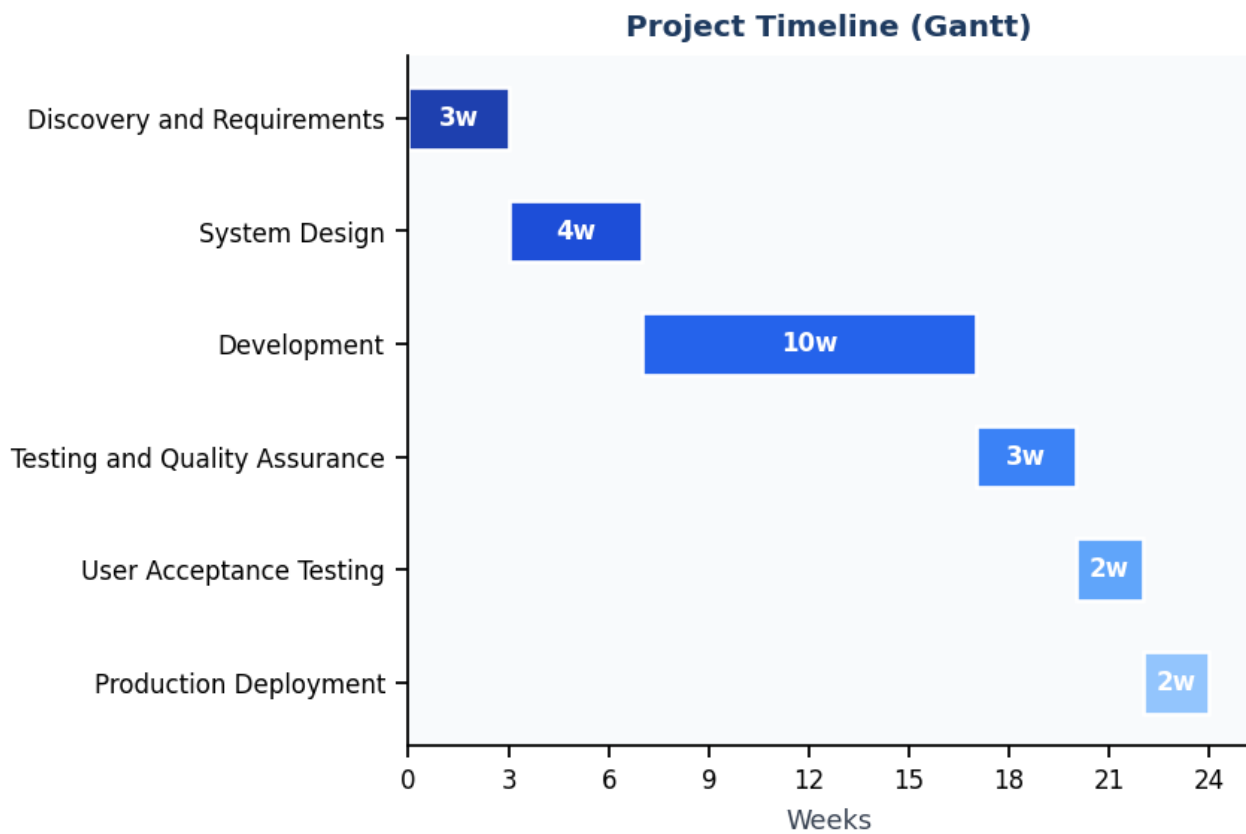
The frontend will be built with Next.js 14+ using App Router architecture, providing a HIPAA-compliant patient portal with role-based access control and real-time result visualization through interactive dashboards. Backend services will leverage Python FastAPI for high-performance RESTful API endpoints, with Scikit-learn and TensorFlow serving the diabetes prediction ML models via a model registry architecture. For the 500-user concurrent workload, a PostgreSQL database will store structured patient records and assessment history, while Redis caching will ensure sub-100ms response times for frequently accessed risk profiles. Deployment will utilize containerized microservices on AWS EKS with GitHub Actions CI/CD pipelines implementing security scanning and automated compliance validation. Monitoring will be provided by Prometheus metrics and Grafana dashboards with PagerDuty alerting, while security posture includes end-to-end encryption, OAuth 2.0 authentication, and comprehensive audit logging for regulatory compliance.

4. Key Deliverables

#	Deliverable
1.	Deployed Production System
2.	API Documentation
3.	User Training Guide
4.	Technical Architecture Document
5.	ML Model Documentation
6.	Security and Compliance Report

5. Project Timeline

Phase	Dur.	Description
Discovery and Requirements	3w	Conduct stakeholder interviews, analyze existing clinical workflows, and define functional requirements resulting in a signed requirements specification document
System Design	4w	Create technical architecture, design database schema, define API contracts, and finalize ML model specifications with architecture decision records
Development	10w	Implement frontend components, build backend APIs, develop ML prediction models, and integrate third-party health data connectors with iterative sprints
Testing and Quality Assurance	3w	Execute unit tests, integration tests, security penetration testing, and ML model validation against clinical accuracy thresholds
User Acceptance Testing	2w	Conduct clinical user validation, perform accessibility testing, and complete HIPAA compliance verification with sign-off documentation
Production Deployment	2w	Deploy to production environment, configure monitoring, execute disaster recovery drills, and transition to operational support with runbooks



6. Estimated Cost Breakdown

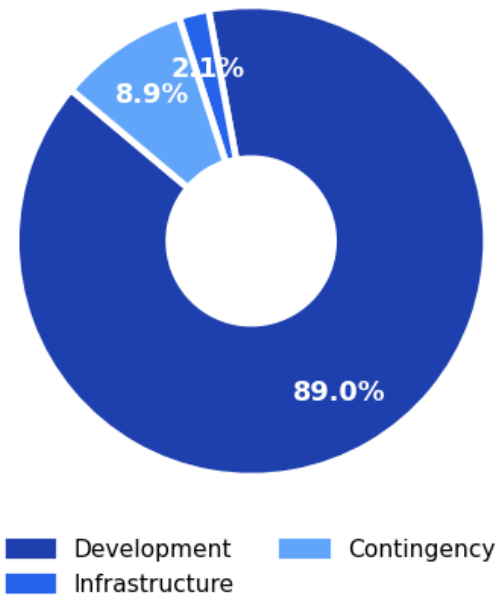
Cost Item	Amount (INR)
Development Cost	Rs. 900,000.00
Infrastructure Cost	Rs. 21,000.00
Contingency (10%)	Rs. 90,000.00
TOTAL ESTIMATED COST	Rs. 1,011,000.00

Monthly average: Rs. 168,500.00

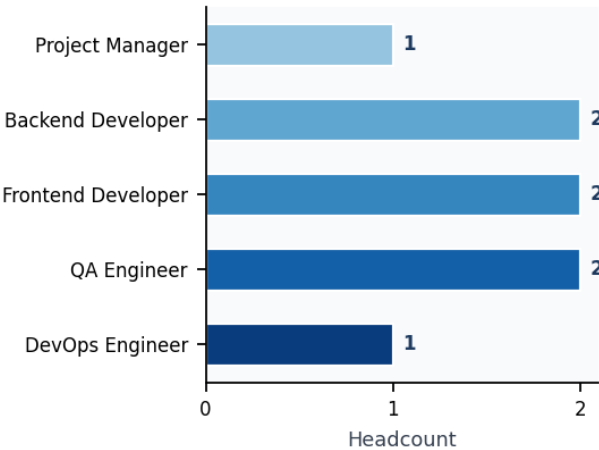
7. Recommended Team Composition

Role	Headcount	Allocation
Project Manager	1	Full-time
Backend Developer	2	Full-time
Frontend Developer	2	Full-time
QA Engineer	2	Full-time
DevOps Engineer	1	Part-time

Cost Distribution



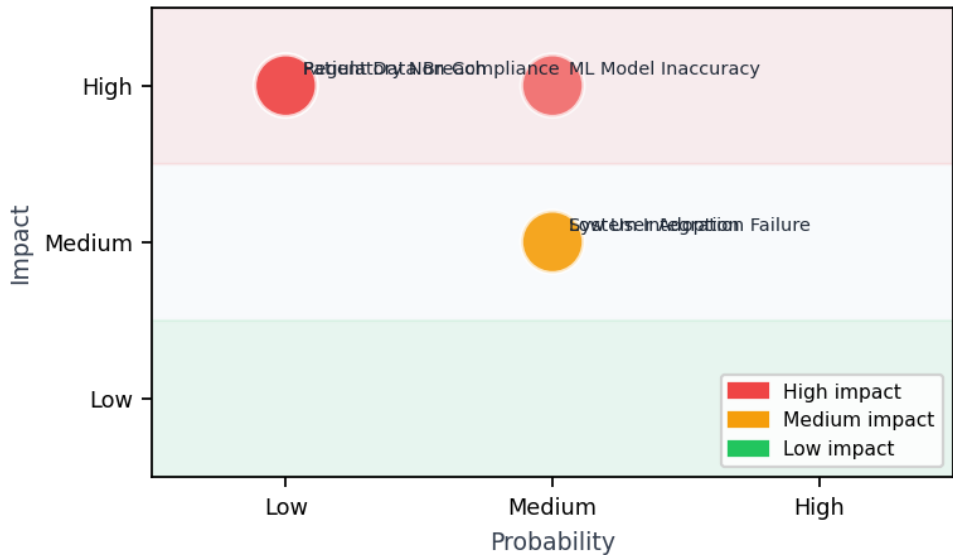
Team Composition



8. Risk Assessment

Risk	Impact	Probability	Mitigation
Patient Data Breach	High	Low	Implement encryption at rest and in transit, enforce principle of least privilege access, and conduct quarterly security audits with vulnerability scanning
ML Model Inaccuracy	High	Medium	Establish continuous model monitoring with drift detection, maintain human-in-the-loop validation for critical decisions, and retrain models with updated clinical data quarterly
Regulatory Non-Compliance	High	Low	Engage compliance consultants for HIPAA gap analysis, implement automated compliance checks in CI/CD pipeline, and maintain audit trails for all PHI access events
System Integration Failure	Medium	Medium	Design with event-driven architecture for loose coupling, implement comprehensive error handling, and maintain fallback procedures for EHR connectivity outages
Low User Adoption	Medium	Medium	Involve clinical end-users in design phases, provide hands-on training sessions, and establish super-user champion network to drive internal adoption

Risk Matrix



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